

STORYTELLING FROM IMAGES

AN EXPLORATION OF IMAGE
TO STORY GENERATION USING
DEEP LEARNING

ANNIE HERRING
ROBERT DALY
DESHANTE HATTORI-LINDSEY
DL DAY 2025

GOAL: GENERATE SHORT STORIES
FROM INPUT IMAGES

GOAL: GENERATE SHORT STORIES FROM INPUT IMAGES

WHY?

1. COMBINES VISUAL AND VERBAL INFORMATION TO STRENGTHEN MEMORY ENCODING AND RECOGNITION
2. EDUCATIONAL APPLICATIONS FOR YOUNG READERS AND LANGUAGE LEARNERS
3. EXPERIMENT WITH COMBINING MULTIPLE MODELS

METHODOLOGY

IMAGE TO PROMPT MODEL

- INPUT: IMAGE
- OUTPUT: CAPTION/PROMPT CORRESPONDING TO IMAGE

PROMPT TO STORY MODEL

- INPUT: PROMPT
- OUTPUT: SHORT STORY BASED ON INPUT PROMPT

METHODOLOGY

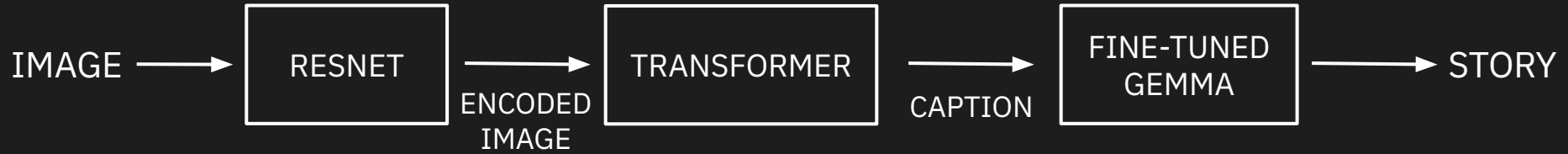


IMAGE TO PROMPT MODEL

IMAGE TO PROMPT MODEL

DATASET:

- FLICKR8K DATASET
- PREPROCESSING:
 - ◆ IMAGES: RESIZING, NORMALIZATION, AND RESNET ENCODING
 - ◆ CAPTIONS: TOKENIZATION, REMOVE INFREQUENT WORDS (<20 OCCURRENCES)

IMAGE TO PROMPT MODEL

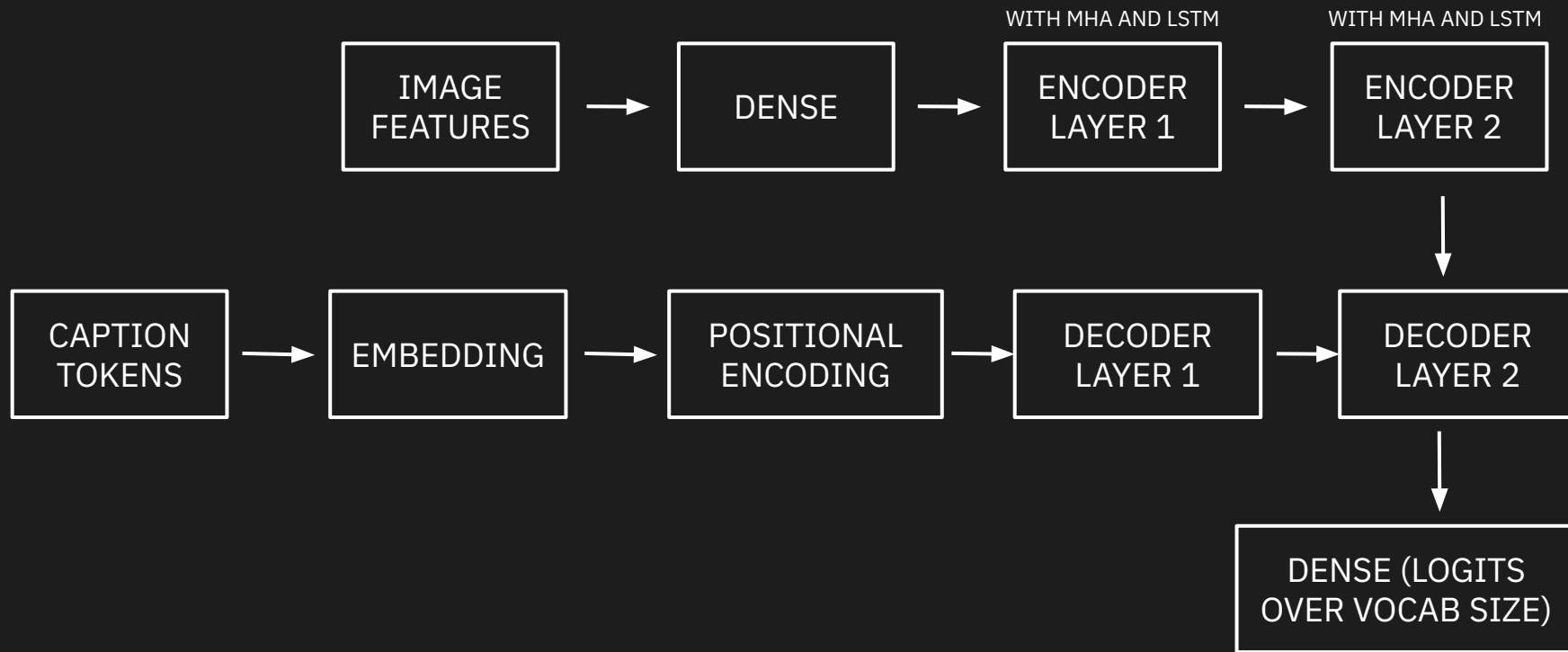


IMAGE TO PROMPT MODEL

TRAINING:

- USED H100 GPU (ROBERT TO EXPLAIN MORE) (JEFF BEZOS DOESN'T LIKE US)
- 20 EPOCHS WITH EARLY STOPPING (IF 4 EPOCHS AND NO IMPROVEMENT)

IMAGE TO PROMPT MODEL

RESULTS: HONESTLY, NOT GREAT



car yard uniforms puppy **climb head** head
dance trail uniforms throws people day
throws snowboarding throws
snowboarding chasing gym past



frisbee picture off face jersey jersey jersey gym
gym gym gym gym **chases animal** soccer gym
gym gym gym rocky

IMAGE TO PROMPT MODEL

WHAT WE TRIED

- DIFFERENT VOCAB SIZE
- ADDING/REMOVING LSTM
- LEARNABLE POSITIONAL ENCODING
- DIFFERENT NUMBER OF ENCODER/DECODER LAYERS

IMAGE TO PROMPT MODEL

WHAT WENT WRONG?

- DATA
- LOSS IS HARD
- VOCAB SIZE

IMAGE TO PROMPT MODEL

HOW COULD WE IMPROVE?

- BETTER IMAGE ENCODING – CLIP, VIT
- VISUAL ATTENTION
- BIGGER/BETTER DATA
- BEAM SEARCH FOR CAPTION GENERATION

FROM-SCRATCH LLM

FROM-SCRATCH LLM

RESULTS: HONESTLY, NOT GREAT

- DATASET: 200K PROMPT TO STORY PAIRS WEB SCRAPPED FROM REDDIT
- SEQ2SEQ TRANSFORMER MODEL
- ARCHITECTURE: EMBEDDING LAYER, POSITIONAL ENCODING, MULTI-LAYER ENCODER-DECODER, AND LINEAR OUTPUT LAYER
- OUTPUTS WERE NONSENSICAL AND WORDS WERE OUT OF ORDER
- MASKING ISSUES: CASUAL MASKING AND PADDING MASKING

FROM-SCRATCH LLM

PROMPT: you walk a lonely road

STORY: you walk a lonely road <newline> , . <newline> of her a his , a <newline> `` , i it i to i i . . in i had ,
and but , of i the `` i and " the the <newline> . the on . . the i the not to . my he <newline> . , he . you my , it
. the i <newline> the at i i the my . me . his that . he to <newline> ? that and . and <newline> the me ``
have " the , . . of ? and . to to at that was a her . his ? . <newline> <newline> it . . . was a i they have the a
, <STOP> the `` i to of i <newline> the his they . to the was . the at <newline> at <newline> <UNK> not a it
that to as the 's in his of he and . this . in on to " of the his <UNK> for , i and i to as <newline> i it , of of . ,
to 's . you of <newline> and ? <STOP> it ? . . <STOP> this was and <newline> <newline> , ? not you . and
for the . you a the `` . that had a and for , . of of , <UNK> have and <newline> the of <UNK> . this in to .
had to to had my a i for .

FINE-TUNING GEMMA

FINE-TUNING GEMMA

PULL THE GEMMA 1B WEIGHTS FROM HUGGING FACE

- INITIAL SIZE: ABOUT 2GB COMPRESSED
- ACTUAL SIZE: ABOUT 4GB UNCOMPRESSED
- ARCHITECTURE: DECODER-ONLY, ATTENTION BASED MODEL. TRAINED ON 2 TRILLION TOKENS WITH VOCAB SIZE OF 262K
- LIGHTWEIGHT: CAN BE RUN ON AN APPLE SILICON M3 CHIP
- EXAMINE OUTPUTS: CALL THE PLAIN MODEL ON PROMPTS FROM OUR TEST SET.

TRAINING

- DATASET: 200K PROMPT TO STORY PAIRS WEB SCRAPPED FROM REDDIT.
- SIZE: APPROXIMATELY 2GB UNCOMPRESSED
- INITIAL PREPROCESSING: THE DATASET IS TOKENIZED FOR EASY SPLITTING ON WHITESPACE.
 - > E.G.: HELLO , THIS IS N'T GREAT !WE REFORMAT TO PLAIN TEXT USING REGEX, WHICH IS THEN SENT TO THE GEMMA TOKENIZER.
- CHATBOT PREPROCESSING: FORMAT AS (SYSTEM, USER, BOT)

TRAINING

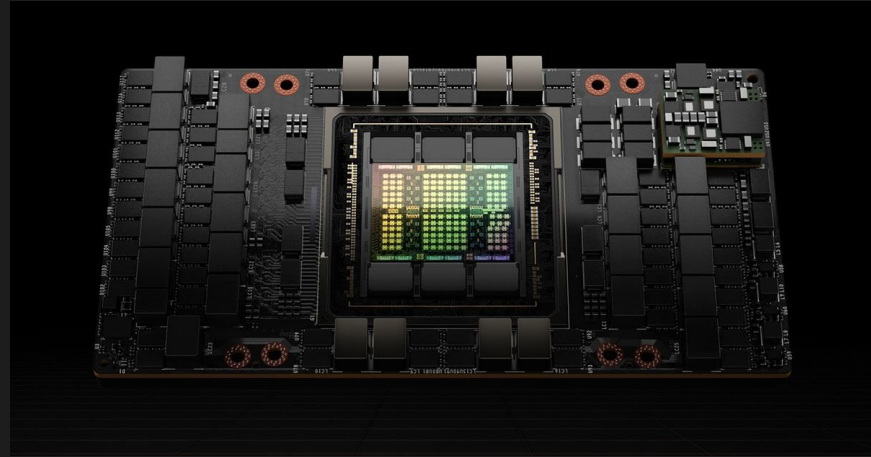
FINE-TUNING ARCHITECTURE:

- USED THE HUGGINGFACE ECOSYSTEM
- IN PARTICULAR, LORA AND PEFT (PARAMETER EFFICIENT FINE TUNING)
- THIS CREATES A TRAINABLE RESIDUAL ADAPTER, BASICALLY ΔW_{LORA} WHICH IS ADDED TO W , WHERE W IS A CHOSEN CLASS WEIGHTS IN THE MODEL.
- THE ADAPTER IS SMALLER 500MB AND MEANS WE DON'T HAVE TO EDIT THE WHOLE MODEL.
- SOME KEY HYPERPARAMETERS: BATCH_SIZE: 4, LEARNING RATE $2E-4$, $R = 4$.

TRAINING

REQUIRES A GPU:

- JEFF BEZOS DOESN'T TRUST ME.
- RAN ON LAMBDA LABS.
- USED AN H100. IT WAS FAST.
- 96 GB VRAM.
- RAN OUT OF MEMORY.
- USED A SUBSET OF DATA.
- RAN OUT OF MEMORY.
- USED A SMALLER BATCH SIZE.
- 11 HOUR RUNTIME.
- TRAINING LOSS FROM 4.637 TO 2.5624.



PROMPT: SEASON 30 OF GAME OF THRONES

GEMMA PLAIN:

The rain is relentless, a grey curtain clinging to the landscape. It's been three years since Jon Snow was slain, and the Iron Throne remains a monument to shattered hope. The North is fractured, ruled by warring lords – the Wildlings, the House Stark, and the Greyjoys – all vying for control of the dwindling resources. Winterfell is a fortress, but it's increasingly isolated, haunted by the ghosts of its past.

FINE-TUNED GEMMA:

I just ca n't believe this is happening." "You think what's happening is happening?" "I'm telling you, I'm telling you! I mean, it's happened before. That's what happened with the White Walkers in the North, that's what happened to the North and it's happened before with the Northern lands. You think I'm stupid? I'm telling you, this is what happens when the White Walkers show up, and it's what happens when they show up." "That's the point, we're supposed to be telling a story, not getting into some **fantasy shit**." "You do n't understand. It's like... like the Wall of the North. The Wall of the North went up in the last episode and it's been like this ever since. And it's not even just the North anymore. It's been going on for a while, and it's not just the North anymore. It's all over the kingdom. Every town in the kingdom is under siege. Every town, every village, every one of them is under siege.

INTEGRATION

RESULTS: HONESTLY, NOT GREAT



"Do n't be such a party animal! I've got to get down to the car yard. The car yard uniform is nice, I'll see you tomorrow."

I stared at her as she walked to the door. I was wearing a full uniform from the car yard, I had to get my uniform cleaned out and I was n't going to be caught short. I had a little trip to the gym to do, and a little party to be at. I threw my arms out to grab her, I do n't know what I would do without her.

Our caption: car yard uniforms puppy
climb head head dance trail uniforms
throws people day throws snowboarding
throws snowboarding chasing **gym** past

INTEGRATION

RESULTS: HONESTLY, NOT GREAT



Ground truth caption: woman climbing in helmet

"And I'm telling you, the rocks are too smooth for me to grip onto, and the wind is too strong for me to hold onto."

"Then why are you here?"

"You're just curious, are n't you?"

"Why?"

"You're curious. That's what you were when you decided to go rock climbing."

"You're not even going to look at me?"

"Yes, I am."

"Why are you so excited about this?"

INTEGRATION

RESULTS: HONESTLY, KIND OF GREAT



Our caption: **frisbee** picture off face jersey
jersey jersey gym gym gym gym gym **chases**
animal soccer gym gym gym gym rocky

Rocky was a strong dog. He was a great dog. He was a good dog.
He was a great dog.

His fur was thick and fluffy, and his eyes were bright and black. He
loved to chase.

Rocky loved the frisbee. He loved the way it flew over the field and
bounced off the grass, and the way it smelled.

Rocky loved to run and run and run. He was the best runner in the
whole wide world. He was the best dog in the whole wide world.

Rocky loved to chase. He loved to chase.

RESULTS

- IN METRICS, 100% PREFERENCE FOR PLAIN GEMMA OUTPUT AS OPPOSED TO FINE-TUNED.
- SOME PLAIN GEMMA OUTPUT WERE PREFERRED TO THE STORIES IN THE TEST SET. GENERAL PREFERENCE FOR HUMAN WRITTEN STORIES STILL.
- IN GENERAL, WE FOUND THAT $\text{QUALITY_LINKED_MODELS} = \text{MIN}(\text{QUALITY_A}, \text{QUALITY_B})$

WHAT COULD WE IMPROVE?

- ADJUST LEARNING RATE
- IMPROVE PREPROCESSING – TOKENIZATION OF CONTRACTIONS
- GEMINI TRAINED FOR NATURAL LANGUAGE – ALREADY GREAT AT MAKING SMALL STORIES
- OUR FINE-TUNING IS AN EXAMPLE OF CATASTROPHIC FORGETTING
- IDEAL USE CASES FOR FINE-TUNING: JAILBREAKING MODELS, HIGHLY SPECIFIC TASK WITH INFORMATION NOT ALREADY IN THE DATASET

DISCUSSION

- TRAINING TWO SEPARATE MODELS IS HARD, CONNECTING THEM IS HARDER
- MORE ADVANCED CAPTIONING TECHNIQUES AND IMPROVING THE OUTPUTS SO BETTER STORY OUTPUTS ARE GENERATED
- FURTHER INTEGRATION OF THE TWO MODELS
- IMPROVE FROM-SCRATCH STORY GENERATOR
 - ◆ HIERARCHICAL NEURAL STORY GENERATION (FAN ET AL., 2018)